



CEN207 Data Structures

Syllabus — Fall Semester, 2026-2027

Asst. Prof. Dr. Uğur CORUH

Course at a glance

- **CEN207 Data Structures** (formerly CE205)
- Compulsory course, 3rd semester
- Theory 3 h/week · Credit 3 · ECTS 5
- Language of instruction: **English**
- Prerequisite: **CEN108 Algorithms and Programming II** (former code CE100)

Course information — contact & logistics

Instructor	Asst. Prof. Dr. Uğur CORUH
Contact	ugur.coruh@erdogan.edu.tr — subject line must start with [CEN207]
Office	F-301
Office hours	By appointment by e-mail; in the office or online with the university account
Lecture day, time, room	Friday 13:00–16:00 · İİBF & Faculty of Law Building, D-402 (ED-K4-2)

Course information — course details

Course website	https://ucoruh.github.io/ce205-data-structures/
Course class	A new class is opened every term; the class code is announced in week 1
Type / semester	Compulsory · 3rd semester
Weekly hours / credit / ECTS	Theory 3 h · Credit 3 · ECTS 5
Prerequisite	CEN108 Algorithms and Programming II (former code CE100)



A. Course Description

What this course is about

- Fundamentals of **data structures** and **file organization**
- How data is mapped in programs: application run-time memory **and** long-term file storage
- Implementations, programming styles and run-time representations of these data objects
- Also covers **sorting**, **searching** and **graph algorithms**

How the course is taught

- Practice-oriented: programming practice in class **and** a term project
- Not theory alone — you build every structure yourself
- Goal: show how digital data structures solve real-world problems

B. Course Learning Outcomes

Learning outcomes (LO.1–LO.7)

Code	Learning outcome
LO.1	Explain definitions, representations and basic operations of linear and non-linear data structures
LO.2	Analyze time/space complexity (Big-O); compare data structure performance
LO.3	Implement fundamental sorting and searching algorithms; analyze and compare them
LO.4	Implement balanced/unbalanced tree structures and hash tables
LO.5	Implement the graph data structure and fundamental graph algorithms
LO.6	Explain sequential, direct and indexed file organization; evaluate applications
LO.7	Select the data structures/algorithms best suited to a problem and build an efficient solution

Contribution of LOs to program outcomes (0–5)

	PO.1	PO.2	PO.3	PO.4	PO.5	PO.6–12
LO.1	5	–	–	–	–	–
LO.2	5	–	–	–	–	–
LO.3	–	–	–	5	–	–
LO.4	–	4	3	5	–	–
LO.5	–	4	3	5	–	–
LO.6	–	–	–	4	5	–
LO.7	5	4	–	3	–	–

PO.1 Basic knowledge · **PO.2** Problem solving · **PO.3** Design · **PO.4** Modern tools/techniques · **PO.5** Research and experimentation



C. Weekly Schedule

The one rule for every assessment

Project demonstrations take place in the week just **before** the midterm and final exam weeks.

Quizzes take place **inside** the midterm and final exam weeks — so that every student can attend.

Weekly schedule (1–8)

Wk	Date	Topic	LO
1	18.09.2026 (make-up 23.09)	Introduction, Big-O, Pointers	1, 2, 7
2	25.09.2026	Linked Lists, Arrays, Matrices	1, 7
3	02.10.2026	Stacks and Queues	1, 7
4	09.10.2026	Trees, Heaps, Huffman	1, 4, 7
5	16.10.2026	Graphs and Traversals	1, 5, 7
6	23.10.2026	Search and Hashing	3, 4, 7
7	30.10.2026	Midterm Project Demonstrations	1–5, 7
8	31.10–08.11.2026	Midterm Exam Week: Quiz-1	1, 2, 4, 5, 7

Weekly schedule (9–16)

Wk	Date	Topic	LO
9	13.11.2026	Graph Algorithms	3, 5, 7
10	20.11.2026	Sorting	2, 3, 7
11	27.11.2026	Advanced Trees	4, 7
12	04.12.2026	Strings	1, 3, 4, 7
13	11.12.2026	File Organisation I	6, 7
14	18.12.2026	File Organisation II	3, 4, 6, 7
15	25.12.2026	Final Project Demonstrations	1–7
16	04–17.01.2027	Final Exam Period: Quiz-2	2–7

Enrichment topics (optional reading)

Kept from previous years, covered in the course notes as **optional** reading:

alpha-beta pruning · Hasse diagrams · Petri nets · bipartite graphs ·
Bayesian networks · van Emde Boas trees · SimHash · trie hashing

D. Textbooks, Software and Equipment

The main resource

The course notes on the course website are self-contained and are the main resource for this course.

The books on the next slide are recommended for further reading only.

Recommended textbooks

- Deitel & Deitel. *C How to Program*, 7th ed. Prentice Hall, 2013.
- Y. Daniel Liang. *Introduction to Java Programming, Comprehensive Version*, 10th ed.
- T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein. *Introduction to Algorithms*, 3rd ed. MIT Press.
- J. R. Hanly, E. B. Koffman. *Problem Solving and Program Design in C*, 6th ed.
- A. L. Tharp. *File Organization and Processing*. Wiley, 1988.
- P. Brass. *Advanced Data Structures*. Cambridge University Press, 2008.
- R. Sedgewick, K. Wayne. *Algorithms*, 4th ed. Addison-Wesley, 2011.

Laptop and toolchain — required

You use your **own development environment** in class, assignments and the project:

- A C/C++ compiler: **GCC**, **Clang** or **MSVC**
- **CMake**, **GoogleTest**, **Doxygen**
- **JDK 21** with **Maven** and **JUnit 5**
- **Git** with a **GitHub** account

Installation steps: week 1 and the course notes. Exact checklist: the course website's **Prerequisites** page.

E. Assessment

One term project, two checkpoints

- One term project, evaluated at **two checkpoints**, each with its own rubric
- **Midterm checkpoint**: C implementation
- **Final checkpoint**: Java implementation
- Plus **one quiz** in the midterm exam week, **one quiz** in the final exam period
- Topics, team rules, deliverables and detailed rubrics: the **project guide**

Assessment components

Assessment	Code	Weight	When
Project checkpoint 1 — C, report, demo	RAP1	60% of midterm	Week 7 (30.10.2026)
Quiz-1 (weeks 1–6)	QUIZ1	40% of midterm	Week 8 (31.10–08.11.2026)
Project checkpoint 2 — Java, report, demo	RAP2	70% of final	Week 15 (25.12.2026)
Quiz-2 (weeks 9–14)	QUIZ2	30% of final	Week 16 (04–17.01.2027)

How the final grade is computed

$$Grade_{Midterm} = 0.6 \cdot RAP1 + 0.4 \cdot QUIZ1$$

$$Grade_{Final} = 0.7 \cdot RAP2 + 0.3 \cdot QUIZ2$$

$$Passing\ Grade = 0.4 \cdot Grade_{Midterm} + 0.6 \cdot Grade_{Final}$$

Workload (ECTS 5 = 125 hours)

Activity	Count	Hours	Total
Class attendance	14	3	42
Individual study (weekly notes/examples)	14	1	14
Quiz (midterm week, final period)	2	2	4
Individual study for quizzes	2	10	20
Project preparation (C and Java)	2	16	32
Report preparation	2	5	10
Project presentation (demo, questions)	2	1.5	3
Total			125



F–K. Policies

F. Instructional strategies and methods

- Lectures are **face-to-face**: explanation, question–answer, hands-on programming
- Each content week: course notes, slides, worked examples, self-check questions
- Announcements, resources and submissions: the course class
- **Attendance is taken**

G. Late homework

- Assignments must be submitted by the **announced deadline**
- **Overdue assignments will not be accepted**
- Unexpected situations must be reported to the instructor **by the student**

H. Course platform and communication

- All announcements, resources and submissions: the **course class**
- A **new class** opens every term; the code is announced in week 1
- Course notes, slides and downloadable documents: the **course website**
- Check the class and your **university e-mail every day**

I. Academic integrity, plagiarism & cheating

Academic integrity is one of RTEÜ's most important principles.

Anyone who breaches it is **severely punished**.

"Studying together" is natural and encouraged — the question is where it crosses into "academic dishonesty." The next two slides draw that line.

I.a What is acceptable

- Discussing the assignment with classmates to understand it better
- Using ideas/quotes/snippets found online **with citation**, not as the whole solution
- Asking for help with the **English** of your assignment
- Discussing solutions with diagrams or summaries, **not actual text or code**
- Working with a tutor, provided the tutor does not do the assignment for you

I.b What is not acceptable

- Asking to see a classmate's solution before submitting your own
- Failing to cite the origin of outside text or code used in your work
- Giving or showing your own solution to a struggling classmate

J. Expectations

- Attend classes **on time**; complete weekly requirements (readings, project milestones)
- Main channel: **e-mail**, from your **university address**
- **Include the course code** in the subject line and **your name** in the message
- The instructor also reaches you by e-mail — check it **every day**

K. Lecture content and syllabus updates

If deemed necessary, the lecture content or course schedule **may change**.

Any change within the scope of this document will be **announced by the instructor**.

Questions?

Course website: ucoruh.github.io/ce205-data-structures

Contact: ugur.coruh@erdogan.edu.tr — subject **[CEN207]**